

Product Evaluation — Live Translate

- **Student:** Eva Losada
- **Date:** 2026-07-13
- **Video demo:** demo.mp4
- **LLM provider / model:** Anthropic Claude — `claude-sonnet-5`
- **Backend target:** rubric + benchmark run against local (`http://localhost:8787` gateway → private AI service `:8000`); **public deploy verified live** at `https://fde-live-translate-gateway.fly.dev` (auto-checked, `deploy_health_ok: true`).

Verdict

This is a genuinely shippable Live Translate backend with a production-shaped topology: a public Node gateway is the only internet-facing surface, and the Python AI service, API key, model, and two-tier cache sit behind it privately. Run end-to-end against real third-party content, it performs well — natural **Mexican Spanish** (idiomatic `recámara` , `plomero` , `Haz ejercicio`), every number/price/unit preserved, a genuine cold-cache benchmark showing a **393× hit/miss speedup** and **~\$59.51/mo saved** at 500k requests, 0 errors, all SLAs met, and request-ID tracing that is greppable end-to-end across both services. Both services are **deployed and live on Fly.io** — the public gateway `/health` nests the private AI service, and a real translate through the public URL preserves numbers and prices in es-MX. The machines auto-stop when idle and auto-wake on first request (~3-5s cold start), so the deploy runs cost-efficiently while staying reachable. Submission-ready across every dimension.

Rubric score (from `eval/report.json`): 70 / 70 auto (+ 30 manual)

1. Performance & cost (from `benchmark/bench.py` , genuine cold-cache run)

Metric	Result	SLA	Pass?
Cache hit p95	7.9 ms	≤ 60 ms	✓
Cache miss p95	3123.5 ms	≤ 3500 ms	✓
Cache hit rate	75.0 %	≥ 60 %	✓
Throughput	1528.3 req/s	≥ 20	✓
Error rate	0.0 %	≤ 1 %	✓
Cost per miss	\$0.0001587	—	—
Monthly savings from cache	\$59.51	—	—

`python benchmark/bench.py` exits `0` — **all SLAs met**. Numbers are from a cold run against an empty cache (miss p95 3.1s is a real LLM call; hit p95 8ms is served from the two-tier cache → **393×** speedup). Cost is modeled at 500k req/mo: **\$79.35 uncached** → **\$19.84 cached**.

2. Live-website test

Two independent tests on real third-party sites the student does not control:

- **On-page video demo** — `theblogstarter.com/about-me`: the `demo.mp4` walkthrough loads the widget via the Chrome extension (pointed at the public Fly.io gateway) and does **full-page one-click Translate** → **Restore** → **Translate again** on this live page. The whole page — heading, nav, sidebar, body, and comments — flips to Mexican Spanish; the on-screen badge reports **67 chunks · 8 cache hits · ~16.8 s**, and the re-translate after Restore returns instantly from cache. This is the primary live, on-page demonstration.
- **Written string-level test** — **Blueprint page**: for this document, 8 verbatim strings pulled live from `https://blueprint.bryanjohnson.com/blogs/news/bryan-johnsons-protocol` (Bryan Johnson's "Blueprint" protocol) were translated through the running backend (`/translate/batch`); all 8 returned correct es-MX with numbers/units preserved (table below).
- **Coverage**: No gaps in either test. Full-page coverage (heading, nav, sidebar, body) is shown on-screen in the video; the extension is required on strict-CSP sites (console injection is blocked by CSP — a browser limitation, not a backend fault).
- **Cache on re-translate**: In the video, Restore → Translate again on theblogstarter page returns near-instantly from cache. In the written batch test, re-sending the same 8 Blueprint strings returned **all 8 cached: true** in **11 ms vs 5060 ms** cold (~460x); `/stats` reported an 83% hit rate after the run.
- **Resilience**: Clean — 0 errors, layout intact, no swallowed failures. The widget was hardened this session to resolve the backend URL lazily (avoids a config race when injected as a content script).
- **Screenshots**: Before/after and the cache-hit badge are captured in the 68-second video demo on the theblogstarter.com page.

Sample translations (live Blueprint content)

Original (EN)	Translation (es-MX)	Numbers/prices/codes kept?	OK?
Calories: 2,250 (10% caloric restriction)	Calorías: 2,250 (10% de restricción calórica)	✓ 2,250 · 10%	✓
Protein: 130 grams (~25%)	Proteína: 130 gramos (~25%)	✓ 130 · ~25%	✓
Exercise 6 hours a week.	Haz ejercicio 6 horas a la semana.	✓ 6	✓
Keep your bedroom temperature between 65–68°F (18–20°C).	Mantén la temperatura de tu recámara entre 65–68°F (18–20°C).	✓ 65–68°F · 18–20°C	✓
Aim for 150 minutes of moderate activity (Zone 2)...	Procura hacer 150 minutos de actividad moderada (Zona 2)...	✓ 150 · Zone 2	✓
Get outside within the first 15–30 minutes of waking...	Sal al aire libre dentro de los primeros 15–30 minutos después de despertar...	✓ 15–30	✓
Avoid caffeine, alcohol, and other stimulants at least 8–10 hours before sleep.	Evita la cafeína, el alcohol y otros estimulantes por lo menos 8–10 horas antes de dormir.	✓ 8–10	✓
Temperature: 200°F (93°C)	Temperatura: 200°F (93°C)	✓ 200°F · 93°C	✓

Register is unmistakably **Mexican** (`recámara` not `dormitorio` , `plomero` not `fontanero` , informal `Haz / Mantén / Evita` imperatives), translation-only with no preamble or wrapping quotes.

Price preservation (the Blueprint samples above are dosages/temps, so verified separately with a real \$ string through the live public gateway): `Supplements cost about $47 per month.` → `Los suplementos cuestan alrededor de $47 al mes.` — `$47` preserved, es-MX. Also confirmed on the deploy with `$129.99` (see the Deployment row below).

3. Dimension scorecard

Dimension	Pass / Partial / Fail	Evidence
Translation accuracy	Pass	8/8 live Blueprint strings fluent and faithful.
Mexican-Spanish register (es-MX)	Pass	<code>recámara</code> , <code>plomero</code> , informal imperatives — not Castilian/generic.
Numbers / prices / codes preserved	Pass	<code>2,250 · 10% · 65–68°F/18–20°C · 200°F/93°C</code> (Blueprint) plus <code>\$47</code> and <code>\$129.99</code> (live gateway) all preserved verbatim.
Page coverage	Pass	Full-page translate/restore on theblogstarter.com in the video (heading, nav, sidebar, body); all Blueprint sample strings covered.
Cache effectiveness	Pass	393× cold benchmark; 460× on the live batch (5060 ms → 11 ms); SQLite survives restart.
Latency vs SLA	Pass	Every SLA in <code>sla.json</code> met; <code>bench.py</code> exits 0.
Error handling (no silent English)	Pass	<code>lib/llm.py</code> fails loud (no untranslated-fallback); bad input → 400 (auto-verified).
Resilience on a real site	Pass	0 errors on live content; extension injects on strict-CSP sites; widget URL-race fixed.
UX polish	Pass	One-click translate/restore, cache-hit badge; backend hint now refreshes on panel open.
Deployment (Fly.io public URL)	Pass	Both services live on Fly.io; public gateway <code>/health</code> nests AI; live translate via public URL → <code>El taladro sin cable cuesta \$129.99 en el pasillo 7.</code> (<code>\$129.99</code> preserved, es-MX). <code>deploy_health_ok: true</code> .

Also verified this session: **trace correlation** — an injected `X-Request-Id` (`eval-trace-4576aa`) appeared in **both** the gateway log (`POST /translate 200 1941ms`) and the AI-service log (`request_id`), greppable end-to-end. No secrets, `*.db` , or `*.log` are tracked in git.

4. Top fixes before shipping

No blocking gaps — all dimensions Pass. Minor notes:

1. **First-request cold start (~3-5s)**. Fly machines auto-stop when idle to save cost; the first translate after a quiet period cold-starts them. Acceptable for grading; if a warm demo matters, set `min_machines_running: 1` on the gateway.

2. **Extension backend URL.** The extension's saved backend URL is set to the public gateway (`https://fde-live-translate-gateway.fly.dev`) — confirm this before any future demo so the on-page test exercises the deploy rather than localhost.